

BADER AL QASEM

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EDUCATION

University of Alberta
Bachelor of Science in Computing Science with Specialization

Edmonton, AB
Expected Dec 2026

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Software Engineering, Database Systems, Programming in C

TECHNICAL SKILLS

Languages: C, Python, Java, JavaScript (ES6+), TypeScript, SQL

Frontend / Backend: React, Node.js, Express, FastAPI, REST APIs, JWT

Databases & Tools: PostgreSQL, MySQL, Docker, Git, GitHub Actions, Azure, Vercel, Render

EXPERIENCE

Full Stack Software Developer
Next Up Hoops (nextuphoops.ca) — [GitHub](https://github.com)

Edmonton, AB
Nov 2024 – Present

- Sole developer for a production web platform supporting a **90+ athlete** basketball organization.
- Designed and deployed a full-stack system using **React, Node.js, Express, and PostgreSQL** on **Vercel and Render**.
- Built secure **JWT-based authentication** and REST APIs supporting user accounts, registrations, and admin workflows.
- Integrated **Stripe Checkout** and automated email flows via **Render** for verification and password recovery.
- Built API test collections in **Postman** to validate authentication, registration, and payment flows.

PROJECTS

Custom Memory Allocator — [GitHub](https://github.com)

- Implemented a custom **malloc/free** allocator in **C** with **heap expansion, block splitting, and coalescing**.
- Designed **dual linked-list** structures (**heap list** and **free list**) for efficient block reuse and memory organization.
- Built comprehensive **allocation, coalescing, and randomized stress tests** to verify allocator correctness

AI Resume Skill Gap Analyzer — [Skill Gap Analyzer](https://github.com) — [GitHub](https://github.com)

- Built an AI-powered service using the **OpenAI API** and **Sentence-BERT** to match resumes with the selected job descriptions.
- Deployed containerized **FastAPI** services to **Azure** with automated **CI/CD pipelines** using GitHub Actions.

Event Lottery App — [GitHub](https://github.com)

- Collaborated in a **6-person Agile team** to develop an **Android application in Java** using **Firebase** that supports Entrant, Organizer, and Admin roles.
- Followed **GitHub workflows** including user-story-driven issues, code reviews, and sprint planning via GitHub Projects.
- Implemented **JUnit** unit tests and **Espresso** UI tests to validate user flows across Entrant, Organizer, and Admin roles.